

# SmartThings Log and Driver Analysis

<https://gemini.google.com/app/d49479f2e9f628ca>

User prompt: look at this CSV file - the author column is associated with a device which has a deviceId. Look for author items of jshimota in the log. Review the log and seek areas where drivers can be worked on to cut down on chatter, poorly formatted text, better explanation and any anomalies.

Response: An analysis of the **24-hour log file** (24hr\_log.txt) matched against your device mapping file (device\_list\_with\_author-09-23-26.csv) was conducted to identify devices and SmartApps authored by **jshimota**.

Below is a breakdown of identified devices, code patterns, chatter sources, errors, and formatting issues, followed by concrete refactoring proposals.

## 1. Authorship Map (jshimota)

The following devices and applications in the log are generated by drivers/apps authored by jshimota:

| ID       | Type   | Name                                                                      |
|----------|--------|---------------------------------------------------------------------------|
| dev:510  | Device | Living Room Thermostat w/Temp (Centralite Pearl Thermostat Custom Driver) |
| dev:1030 | Device | AVT - Office Thermostat (Custom) (Virtual Thermostat Driver)              |
| dev:1049 | Device | WOL - Titbeast2 (Wake-on-LAN Driver)                                      |
| dev:1024 | Device | Virtual Mode Status Tracker Tile (Dashboard Tile Driver)                  |
| dev:1025 | Device | Alexa Mode - Voice Source (Virtual Switch)                                |

| ID       | Type   | Name                                         |
|----------|--------|----------------------------------------------|
| dev:1008 | Device | Alexa Mode - Awake ( <i>Virtual Switch</i> ) |
| app:922  | App    | Multiperiod Environment Manager v1.8.6       |
| app:937  | App    | Mode Manager Advanced                        |
| app:943  | App    | Advanced Porch Lighting Manager v3.4.1       |
| app:946  | App    | Echos Speak Advanced v2.1.7                  |

## 2. Analysis of Log Chatter, Errors, and Formatting Issues

### A. Centralite Pearl Thermostat Driver (dev:510) — *High Chatter & Event Storming*

- Thermostat Setpoint Flipping / Ping-Ponging:** During scheduled setpoint adjustments or temperature updates, dev:510 generates excessive log entries due to setpoint bouncing back and forth within milliseconds:
  - 08:52:10.948 AM info Living Room Thermostat w/Temp - thermostatSetpoint was set to 74 °F
  - 08:52:10.888 AM info Living Room Thermostat w/Temp - thermostatSetpoint was set to 70 °F
  - 08:42:10.990 AM info Living Room Thermostat w/Temp - thermostatSetpoint was set to 74 °F
  - 08:42:10.900 AM info Living Room Thermostat w/Temp - thermostatSetpoint was set to 70 °F
- Redundant Command and Event Execution:** During mode transitions (e.g., transition to Morning at 08:37:08 AM), dev:510 logs 14 lines in under **0.5 seconds**:
  - 08:37:09.098 AM info ... coolingSetpoint was set to 74 °F
  - 08:37:09.096 AM info ... Setting coolingSetpoint to 74 °F
  - 08:37:08.819 AM info ... thermostatSetpoint was set to 74 °F
  - 08:37:08.760 AM info ... thermostatSetpoint was set to 74 °F
  - 08:37:08.733 AM info ... coolingSetpoint was set to 74 °F
  - 08:37:08.731 AM info ... Setting coolingSetpoint to 74 °F
  - 08:37:08.619 AM info ... coolingSetpoint was set to 74 °F
  - 08:37:08.617 AM info ... Setting coolingSetpoint to 74 °F
  - 08:37:08.523 AM info ... thermostatSetpoint was set to 70 °F
  - 08:37:08.510 AM info ... heatingSetpoint was set to 70 °F
  - 08:37:08.508 AM info ... Setting heatingSetpoint to 70 °F
  - 08:37:08.503 AM info ... thermostatSetpoint was set to 70 °F
  - 08:37:08.377 AM info ... heatingSetpoint was set to 70 °F
  - 08:37:08.375 AM info ... Setting heatingSetpoint to 70 °F

*Key Issue:* The driver logs both Setting heatingSetpoint to X (command logger) and heatingSetpoint was set to X (event logger). When dual-setpoint or auto-mode setpoint calculations trigger, internal setter methods are executed multiple times concurrently without state debouncing.

## B. Echoes Speak Advanced (app:946) & Echo Devices — *Unhandled Network Timeout Exceptions*

- **Bridge Communication Failures:** When the Node.js bridge server on 192.168.1.12:4000 is unresponsive or drops a request, unhandled errors bubble up to system logs:  
08:37:25.905 AM error Guest Bathroom Echo Show 8": Guest Bathroom Echo Show 8" - voiceCmdAsText command failed on bridge server.  
08:37:25.902 AM error Echos Speak Advanced v2.1.7 - [15 Device(s)]: HTTP Post error sending command 'voiceCmdAsText' to device [G0916D100067042U]: 192.168.1.12:4000 failed to respond  
08:37:15.431 AM error Echos Speak Advanced v2.1.7 - [15 Device(s)]: HTTP Post error sending command 'speak' to device [G6G22N063493008N]: 192.168.1.12:4000 failed to respond

*Key Issue:* Network timeouts flood the log with raw socket error strings (failed to respond) without exponential backoff or quiet retry logic.

## C. Multiperiod Environment Manager (app:922) — *Redundant Evaluation Logging*

- During state transitions, app:922 logs duplicate transition messages in rapid succession:  
08:37:09.657 AM info Multiperiod Environment Manager v1.8.6: Period transitioned to Morning | House Target: Heat 70 °F, Cool 74 °F | Office Target: Heat 74 °F, Cool 80 °F  
08:37:09.045 AM info Multiperiod Environment Manager v1.8.6: Period transitioned to Morning | House Target: Heat 70 °F, Cool 74 °F | Office Target: Heat 74 °F, Cool 80 °F  
08:37:08.294 AM info Multiperiod Environment Manager v1.8.6: Period transitioned to Morning | House Target: Heat 70 °F, Cool 74 °F | Office Target: Heat 74 °F, Cool 80 °F  
08:37:08.163 AM info Multiperiod Environment Manager v1.8.6: Period transitioned to Morning | House Target: Heat 70 °F, Cool 74 °F | Office Target: Heat 74 °F, Cool 80 °F

*Key Issue:* Multiple event subscribers (e.g., location mode event, thermostat setpoint change event, switch change event) each trigger full re-evaluations without state guards, printing the exact same log statement 4 times in 1.5 seconds.

## D. Formatting and Typo Anomalies

- **Device dev:1049 (WOL Driver):**
  - Current Log: WOL - Titbeast2: Sent WOL magic packet...
  - Clarification: Verify if Titbeast2 is intentional or a placeholder/typo naming issue.
- **OpenWeatherMap Driver / AQI Driver (dev:217 / dev:224 / dev:909):**
  - Logs state changes every 5–15 minutes with standard text (Polling AirNow..., OpenWeatherMap AQI Tile city is Dundee). Ensure INFO logging is toggled via configurable txtEnable so log space is preserved when debug logging is disabled.

## 3. Actionable Driver & SmartApp Improvement Plan

### Action Item 1: Debounce Setpoint Writes in Centralite Pearl Thermostat Driver (dev:510)

- **Problem:** Internal setpoint commands write to child states and fire events even if the target value matches the current state, causing downstream app chain triggers.
- **Fix:** Add value-comparison checks before sending physical Z-Wave/Zigbee commands or producing events:  

```
void setHeatingSetpoint(degrees) {  
    BigDecimal target = degrees.toBigDecimal()
```

```

BigDecimal current = device.currentValue("heatingSetpoint")?.toBigDecimal()
if (target == current) {
    if (txtEnable) log.debug "${device.displayName}: Heating setpoint already ${target}°F. Skipping
command."
    return
}
if (txtEnable) log.info "${device.displayName}: Setting heating setpoint to ${target}°F"
// Send Zigbee/Z-Wave cluster command...
}

```

## Action Item 2: Suppress Duplicate Log Outputs in Centralite Pearl Thermostat (dev:510)

- **Problem:** Driver prints both "Setting heatingSetpoint to 70 °F" (in the command execution) and "Living Room Thermostat w/Temp - heatingSetpoint was set to 70 °F" (in parse/event generation).
- **Fix:** Remove log statements inside internal command methods when sendEvent(isStateChange: true) handles standard logging during Zigbee/Z-Wave report parsing.

## Action Item 3: Add State Guard / Re-evaluation Debouncing in Multiperiod Environment Manager (app:922)

- **Problem:** Upstream mode changes fire multiple attribute updates, causing app:922 to run evaluatePeriods() 4 times in 1 second.
- **Fix:** Implement a 1–2 second runIn execution buffer (atomic state execution check):

```

def modeOrStateChangeHandler(evt) {
    runIn(1, "evaluatePeriodsDebounce", [overwrite: true])
}

def evaluatePeriodsDebounce() {
    // Perform single evaluation after event cascade settles
}

```

## Action Item 4: Enhance Resilience & Quiet Logging in Echos Speak Advanced (app:946)

- **Problem:** HTTP post timeouts to Node.js server write noisy call stacks to Hubitat logs.
  - **Fix:** Wrap HTTP POST calls in explicit error-catching logic with concise single-line warning logs and automated retry logic:

```

try {
    httpPostJson(params) { response -> /* process */ }
} catch (java.net.SocketTimeoutException e) {
    if (logEnable) log.warn "Echo Bridge (${bridgeIp}): Connection timed out for ${device.displayName}.
Retrying in 5s..."
    runIn(5, "retryCommand", [data: [cmd: command, deviceId: deviceId]])
} catch (Exception e) {
    log.error "Echo Bridge Error [${device.displayName}]: ${e.message}"
}

```
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